

Table 5. Multiple Linear Regression: Final Model Coefficients  
Body Mass Predicted by Morphometric Variables, Species, and Sex

| Parameter             |
|-----------------------|
| Estimate              |
| Std. Error            |
| 95% CI                |
| p-value               |
| Intercept             |
| -1460.99              |
| 571.31                |
| [-2584.9, -337.1]     |
| 0.011                 |
| Flipper length (mm)   |
| 15.95                 |
| 2.91                  |
| [10.2, 21.7]          |
| <0.001                |
| Bill length (mm)      |
| 18.20                 |
| 7.11                  |
| [4.2, 32.2]           |
| 0.011                 |
| Bill depth (mm)       |
| 67.22                 |
| 19.74                 |
| [28.4, 106.1]         |
| <0.001                |
| Species (ref: Adelie) |
| Chinstrap             |
| -251.48               |
| 81.08                 |
| [-411.0, -92.0]       |
| 0.002                 |
| Gentoo                |
| 1014.63               |
| 129.56                |
| [759.7, 1269.5]       |
| <0.001                |

Sex (ref: female)

male

389.89

47.85

[295.8, 484.0]

<0.001

R-squared: 0.875 | Adjusted R-squared: 0.873 | RMSE: 284.3 g

Source: Palmer Penguins Dataset (Gorman et al., 2014)

Dependent variable: Body mass (g)

Reference categories: Species = Adelie; Sex = female